



Exploring the mechanism of *in situ* formation of oxalic acid for producing mixed fumarato-oxalato lanthanide (Eu, Tb and Gd) frameworks

Cristiane K. Oliveira^a, Juliano R. de Menezes Vicenti^b, Robert A. Burrow^b, Severino Alves Jr.^a, Ricardo L. Longo^a, Ivani Malvestiti^{a,*}

^a Departamento de Química Fundamental, UFPE, 50.740-560, Recife, PE, Brazil

^b Departamento de Química, UFSM, 97105-900, Santa Maria, RS, Brazil

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ABSTRACT

Mixed fumarato-oxalato ligands lanthanide metal-organic frameworks (MOFs) $[\text{Ln}_2(\text{fum})_2(\text{ox})(\text{H}_2\text{O})_4] \cdot 4\text{H}_2\text{O}$ ($\text{Ln} = \text{Eu, Gd, and Tb}$) were obtained from hydrothermal treatment of aqueous acid solutions of trivalent lanthanide ions and fumaric acid with nitrate or sodium hypochlorite as oxidants. The crystal structure for $[\text{Gd}_2(\text{fum})_2(\text{ox})(\text{H}_2\text{O})_4] \cdot 4\text{H}_2\text{O}$ was determined. Other oxidants (H_2O_2 and NaSO_4) or conditions (NaOCl at room temperature and atmospheric pressure) yielded other MOFs or products. Oxalic acid was formed *in situ* and mechanisms based on the hydration of fumaric acid mediated by Ln^{3+} followed by oxidation and decarboxylation steps were proposed. The mechanisms were able explain the dual role of Ln^{3+} ions and the nature of the oxidant in forming the frameworks.

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Lanthanide-based coordination polymers and metal-organic frameworks (MOFs) are relevant in materials chemistry due to their tunable adsorption, catalytic, optical, and magnetic properties [1,2]. However, the prediction of the crystalline and/or framework structures is hindered by the flexible and versatile coordination polyhedra of the lanthanide ion. This precludes the formation of a well-defined secondary building unit used in the reticular synthesis approach [3]. Also, some ligands are generated *in situ* and their formation is still unclear, so the structure predictability becomes difficult. In fact, *in situ* formation of ligands is important because their kinetics can control crystal growth and produce mixed ligands new hybrid materials that would otherwise be very difficult, or even impossible, to obtain from a mixture of the ligands [4]. Oxalate *in situ* formation under several conditions is quite widespread and has been used to generate coordination polymers and MOFs with a variety of transition metal, lanthanide and actinide ions [4]. Despite its relevance for preparing new compounds and hybrid materials the mechanism for *in situ* generation of oxalate has not been explored. Thus, we intend to use the *in situ* formation of oxalic acid from a hydrothermal treatment of acid aqueous solution of fumaric acid and trivalent lanthanide nitrate to explore possible pathways and effects for this reaction. Indeed, mixed fumarato (fum) and oxalato (ox) lanthanide porous frameworks $[\text{Ln}_2(\text{fum})_2(\text{ox})(\text{H}_2\text{O})_4] \cdot 4\text{H}_2\text{O}$ ($\text{Ln} = \text{Eu and Tb}$) have already been prepared [5] using fumaric acid

and oxalic acid as starting materials and lanthanide nitrate under hydrothermal conditions. We report the synthesis of $[\text{Ln}_2(\text{fum})_2(\text{ox})(\text{H}_2\text{O})_4] \cdot 4\text{H}_2\text{O}$ ($\text{Ln} = \text{Eu, Gd, and Tb}$) frameworks where, for the first time, oxalate was generated *in situ* from fumaric acid. Considering the simplicity of the reacting system: $\text{Eu}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$, fumaric acid, and water, it seems appropriate its use for exploring the mechanism of *in situ* generation of oxalate.

$\text{Ln}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$ ($\text{Ln} = \text{Eu, Gd, and Tb}$) were prepared by dissolving commercially available analytical grade lanthanide oxide in concentrated nitric acid followed by complete evaporation of the solution. All other starting materials were commercially available reagents of analytical grade and used without further purification. Water was obtained from a milli-Q equipment. $[\text{Eu}_2(\text{fum})_2(\text{ox})(\text{H}_2\text{O})_4] \cdot 4\text{H}_2\text{O}$ framework (**1Eu**) was prepared from a mixture of $\text{Eu}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$ (0.40 mmol), fumaric acid (*E*-butenedioic acid, 1.5 mmol), and H_2O (5 mL) sealed in a Teflon-lined vessel (12 mL) and heated at 160 °C for 12 h. The reaction vessel was allowed to slowly cool at room temperature. Single crystals of **1Eu** were collected by filtration and washed with deionized water and ethanol, and dried in air. Yield: 70% based on Eu^{3+} salt. The initial and final pH were within 1–2. The same procedure was used to prepare $[\text{Gd}_2(\text{fum})_2(\text{ox})(\text{H}_2\text{O})_4] \cdot 4\text{H}_2\text{O}$ (**1Gd**) and $[\text{Tb}_2(\text{fum})_2(\text{ox})(\text{H}_2\text{O})_4] \cdot 4\text{H}_2\text{O}$ (**1Tb**) MOFs by changing the lanthanide nitrate in yields of 65% and 60%, respectively. Elemental analysis in percentages found (calculated) of the compounds were: **1Eu** C = 15.58% (15.72) and H = 3.08% (2.64); **1Tb** C = 15.67% (15.44) and H = 2.76% (2.59).

The three lanthanide frameworks $[\text{Ln}_2(\text{fum})_2(\text{ox})(\text{H}_2\text{O})_4] \cdot 4\text{H}_2\text{O}$ ($\text{Ln} = \text{Eu, Gd, and Tb}$) were analyzed by single crystal and powder X-ray diffraction (PXRD) [6]. The structures of **1Eu** and **1Tb** were

* Corresponding author at: Departamento de Química Fundamental, Universidade Federal de Pernambuco, Av. Jornalista Aníbal Fernandes, s/n - Cidade Universitária, 50.740-560, Recife, PE, Brazil. Tel.: +55 81 21267471; fax: +55 81 21268442.

E-mail address: ivani@ufpe.br (I. Malvestiti).

already fully discussed in the literature [5] and the structure of **1Gd**, which is isomorphous to those of **1Eu** and **1Tb**, is provided in the Supplementary Material (Fig. S1).

The PXRD for **1Eu** (Fig. 1a) and **1Tb** are fully consistent with that available in the literature [5] at distinct degrees of dehydration. We emphasize that no oxalic acid was present in the initial solutions, so the oxalato ligand observed in the structure originated *in situ* from fumaric acid and/or CO₂ (dissolved from the atmosphere or from decarboxylation of fumaric acid).

Control experiments were performed to provide insights into the oxalate formation. The aqueous solution of europium nitrate and fumaric acid was degassed with argon gas for several hours prior to the hydrothermal treatment and gave **1Eu** in good yield (60%). It can thus be deduced that neither O₂(g) nor CO₂(g) were needed for producing oxalic acid. The hydrothermal treatment of 5 mL aqueous solution of europium chloride (0.40 mmol), sodium nitrate (1.2 mmol) and fumaric acid (1.5 mmol) with initial pH 1–2 afforded **1Eu** in good yield (64%), indicating that the source of the nitrate is irrelevant. However, in the absence of nitrate, that is, aqueous solution of europium chloride (0.40 mmol) and fumaric acid (1.5 mmol) at pH 1–2 did not yield **1Eu** under the same hydrothermal conditions, which showed the need of nitrate for oxalate formation from fumaric acid (Figs. S2–S5). Also, the formation of oxalic acid requires the presence of Ln³⁺ ions, since the hydrothermal treatment of 5 mL aqueous solution of sodium nitrate (1.2 mmol) and fumaric acid (1.5 mmol) with pH 1–2 did not yield oxalic acid when the resulting solution was analyzed by NMR (see Figs. S6–S9). Another relevant result is that the iso-structural [Ln₂(fum)₂(ox)(H₂O)₄]·4H₂O frameworks were obtained for Ln = Eu, Gd, and Tb, which have significantly different Ln³⁺ + e[−] → Ln²⁺ redox potentials [7]. We also observed that when aqueous solutions of europium chloride and oxalic acid were mixed a precipitate was immediately obtained. Thus, when aqueous solutions of europium chloride, fumaric acid and oxalic acid were mixed only oxalato-europium containing solid is obtained due to large differences in acidity and solubility. That is, oxalic acid is almost 100 times more acid (pK_{a1} = 1.25 and pK_{a2} = 3.81 at 25 °C) than fumaric acid (pK_{a1} = 3.02 and pK_{a2} = 4.38 at 25 °C) and at least 10 times more water soluble (95.2 mg per mL H₂O at 20 °C) than fumaric acid (6.3 mg per mL H₂O at 25 °C and 99.7 mg per mL H₂O at 100 °C). This shows that the slow formation of oxalic acid during the hydrothermal treatment is essential for obtaining the mixed ligands (fumarato-oxalato) lanthanide MOFs. The reported syntheses [5] of **1Eu** and **1Tb** employed hydrothermal treatment of aqueous solution of europium nitrate, fumaric and oxalic acids and we propose that the hydrothermal conditions were needed

for dissolving the oxalato-lanthanide framework, for increasing the fumaric acid solubility and dissociation so mixed ligand oxalato-fumarato lanthanide crystals could be formed.

From the experimental evidence, the formation of oxalate from reductive coupling of dissolved CO₂ [8] can be discarded. However, it has been proposed that CO₂ produced from decarboxylation of organic (poly)carboxylic acids might be the source of oxalate via reductive coupling. So, it is appropriate to briefly review the *in situ* formation of oxalate, which we have grouped as oxidation processes:

- ox1) oxidation of the squarate by Ce(IV) or hydrogen peroxide at room conditions [9] or of squaric acid in aqueous solution of Cu(CH₃COO)₂·H₂O and NH₃ at 60 °C [10] or of squaric acid in the presence of CoCl₂ and imidazole at pH 8 under hydrothermal conditions [11];
- ox2) hydrolysis of amino- and thio-derivatives of squaric acid in the presence of La³⁺ and Bi³⁺ ions [12];
- ox3) oxidation/ring opening of the methoxy- and methyl-squarate in acetonitrile or aqueous solutions containing *cis*-[RuCl₂(dmsO)₄] under slow evaporation at 28 °C [13];
- ox4) oxidation-hydrolysis under (solvo)hydrothermal conditions of pyridine-4-carboxylic acid in aqueous solution of Zn(NO₃)₂·6H₂O and iodine [14] or of orotic acid (2,6-dioxo-1,2,3,6-tetrahydropyrimidine-4-carboxylic acid) in aqueous solution of Sm(NO₃)₃·6H₂O and 2CoCO₃·3Co(OH)₂·nH₂O [15] and in a mixture of lanthanide oxide, CuCl₂·2H₂O and water [16] or of pyrazine-2-carboxylate in aqueous solution of lanthanide oxide (Pr₆O₁₁ and Er₂O₃ [17] or Nd₂O₃ [18]) and FeCl₃ [17] or Fe(NO₃)₃ [18] or of pyridine-3-carboxylic acid in aqueous solution of lanthanide oxide (Nd₂O₃ and Eu₂O₃) and CuCl₂·2H₂O [19] or of 3-aminopyrazine-2-carboxylic acid in aqueous solution of lanthanide oxide (Pr₂O₃ and Nd₂O₃) [20] or of pyrimidine-4,6-dicarboxylic acid in aqueous solution of lanthanide nitrate or chloride [21] or of pyridine-2,4,6-tricarboxylic acid in ethanolic solution of Cd(NO₃)₂·4H₂O [22] or of 2-cyanopyrimidine in aqueous solution mediated by CdCl₂ [23] or by MnCl₂ [24] or of glycolic acid or of ethylene glycol in aqueous solution of (UO₂)(NO₃)₂·6H₂O and tetraethylammonium hydroxide (pH 2–3) [25] or of iminodiacetic acid in aqueous solution of Ln(ClO₄)₃·6H₂O (Ln = Pr and Nd), Cu(ClO₄)₂·6H₂O and HCl (pH adjusted to 3) [26];
- ox5) oxidation of the ethanol or oxidative coupling of methanol molecules under oxygen-free hydro(solvo)thermal conditions mediated by [Zn(NO₃)₂·6H₂O] and pyridine with the nitrate being suggested as oxidant [27].

The following reduction processes might be grouped as,

- red1) selective reductive coupling of dissolved CO₂ by (C₅Me₅)₂Sm(THF)₂ in dried THF and nitrogen atmosphere [28], with Sm(II) complex acting as an electron source and as a mediator by presumably forming “[(C₅Me₅)₂Sm(THF)_x]⁺[CO₂][−]” species that couples to form oxalato complex;
- red2) reductive coupling of dissolved CO₂ in methanol containing CuI, Ph₄B and a triazacyclononane ligand, with the Cu(I) being the electron source and the active species was the bicarbonate anion [29];
- red3) reductive coupling of dissolved CO₂ mediated by a mixture of alkyltetraamine diol, 2,6-(bis(chloromethyl)phenol and Ln(NO₃)₃·nH₂O (Ln = La, Pr) in nitric acid with nitrate being indispensable, but the electron source was not identified [30];
- red4) reductive coupling of CO₂ from decarboxylation mediated by metal ions under hydrothermal treatment of aqueous solutions of pyridine-3,5-dicarboxylic acid or pyridine-2,4-dicarboxylic acid and Tb(NO₃)₂·5H₂O [31] or of pyridine-3,5-dicarboxylic acid, Ln(NO₃)₂·5H₂O (Ln = Eu, Sm, Ho, and Dy) and 4,4'-bipyridine [32] or of pyridine-3,5-dicarboxylic acid, lanthanide

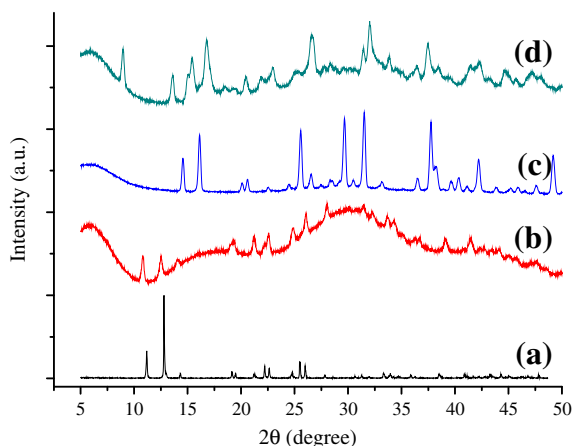


Fig. 1. Powder X-ray diffractograms for the solids obtained from the hydrothermal treatment of aqueous solutions of lanthanide nitrate and fumaric acid (a), of lanthanide chloride, fumaric acid and sodium hypochlorite (b) or sodium sulfate (c), and from an aqueous solution of lanthanide chloride, fumaric acid and sodium hypochlorite at room temperature and atmospheric pressure (d).

oxide and perchloric acid [33] or of $\text{H}_2\text{pda} \cdot 2\text{HCl}$ ($\text{H}_2\text{pda} = 1,4$ -piperazinediacetic acid), Ln_2O_3 ($\text{Ln} = \text{Nd}, \text{Y}, \text{Eu}, \text{CeO}_2$ and Pr_6O_{11}) and 50% H_2SO_4 [34] or of pyridine-3,4-dicarboxylic acid, lanthanide oxide (Pr_6O_{11} and Ln_2O_3 with $\text{Ln} = \text{Nd}, \text{Eu}, \text{Gd}, \text{Tb}$ and Er) and HNO_3 (adjusted pH 2.5) [35] or of pyridine-4-carboxylic acid, fumaric acid, Eu_2O_3 , AgNO_3 and HNO_3 [36] or of pyrazine-2,3-dicarboxylic acid and lanthanide nitrate ($\text{Ln}^{3+} = \text{Ce}, \text{Nd}, \text{Sm}, \text{Eu}, \text{Gd}, \text{Tb}$, or Er) [37] or of 5-hydroxy-benzene-1,3-dicarboxylic acid, $\text{Sm}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$ and NaOH [38] or of 2,5-dihydroxy-1,4-benzenedicarboxylic acid or tetrabromobenzene-1,4-dicarboxylic acid, $\text{Ln}(\text{NO}_3)_3 \cdot n\text{H}_2\text{O}$ ($\text{Ln} = \text{La}$ and Ho , $n = 6$ and 5) and KOH [39].

Some interesting patterns emerge from this grouping, namely, that most oxidation-hydrolysis processes (ox4) involve *N*-heterocyclic or amine (poly)carboxylic acids, whereas the proposed decarboxylation/ CO_2 reductive coupling mechanism (red4) also involves mostly *N*-heterocyclic (poly)carboxylic acids. Additionally, in all these later processes (red4) the aqueous medium is acidic and oxidizing, that is, contains nitrate, perchlorate, and/or sulfate. Indeed, since the initial proposal [31] of this decarboxylation/ CO_2 reductive coupling mechanism for the *in situ* formation of oxalate from *N*-heterocyclic (poly)carboxylic acids under hydrothermal conditions in the presence of metal ions (mainly, trivalent lanthanides), all reported syntheses under similar conditions invoke this mechanism without any evidence for the reductive coupling step or even for the electron source(s). In fact, there is a recent direct evidence for a decarboxylation step during the hydrothermal treatment of 2,3-pyrazinedicarboxylic acid and lanthanide nitrate due to the presence of 2-pyrazinedicarboxylato (2-pzc) in the by-product $[\text{Tb}_2(2\text{-pzc})_4(\text{ox})(\text{H}_2\text{O})_6] \cdot 10\text{H}_2\text{O}$ [37]. This step might also be involved in the hydrothermal treatment of a solution containing pyridine-3,5-dicarboxylic acid, lanthanide nitrate and 4,4'-bipyridine, where dihydroxypyridine ($m/e = 111$) was detected by GC–MS analysis of the resulting solution [32]. One or both carboxylate groups in pyridine-3,5-dicarboxylic acid were replaced by iodide in the presence of $\text{Cu}(\text{II})$ under hydrothermal conditions [40]. These latter evidences are indirect since only the substituted (hydroxy or iodide) products were detected instead of the decarboxylated ones. We also have indirect evidence that this decarboxylation step occurs in the hydrothermal treatment of acid aqueous solutions of fumaric acid and europium nitrate (formation of **1Eu**) or of fumaric acid and europium chloride (absence of **1Eu**). Namely, NMR analyses [41] with addition of known quantities of 3-(trimethylsilyl)propionic-2,2,3,3- d_4 acid sodium salt (TSP) in the solution resulting from latter experiment (absence of **1Eu**) revealed a loss (15% in mass) of fumaric acid (Figs. S2–S9). We propose that under hydrothermal conditions, fumaric acid can be decarboxylated in the presence of Ln^{3+} and the volatile species formed were not detected for the mass balance. The direct decarboxylation of fumaric acid is a highly activated process, so an intermediate hydration step (electrophilic addition) forming 2-hydroxysuccinic acid, according to Eq. (1), is proposed.

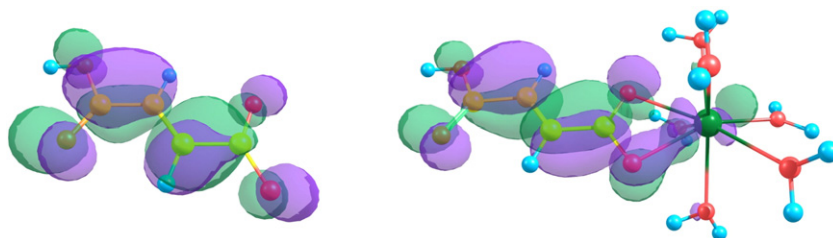


Fig. 2. Lowest unoccupied molecular orbital (LUMO) of mono-ionized fumarate ($\text{fumH}^- = \text{HOOCCH}=\text{CHCOO}^-$) and the $[\text{Ln}(\text{H}_2\text{O})_6 \cdot \text{fumH}]^{2+}$ complex calculated with the B3LYP/MWB52(Eu)/6-31 + G method. Iso-surface value = 0.03 e_0^{-3} .

This α,β -hydroxydicarboxylic acid can be decarboxylated to produce, for instance, 3-hydroxypropionic acid according to Eq. (2).



Both processes (hydration and decarboxylation) are mediated by Ln^{3+} and probably accelerated under hydrothermal conditions. Within the experimental uncertainty of the NMR analyses ($\sim 5\%$) we did not observe any loss of fumaric acid when the same hydrothermal treatment was performed in the absence of Ln^{3+} . Indirect evidence for the formation of 3-hydroxypropionic acid in Eq. (2) is the presence of 3-(pyridine-4-carboxylic acid)propionic acid in the europium MOF obtained from the hydrothermal treatment of aqueous solution of pyridine-4-carboxylic acid, fumaric acid, Eu_2O_3 , AgNO_3 and HNO_3 [36]. Quantum chemical calculations also suggest that the hydration step of the double bond in Eq. (1) is mediated by Ln^{3+} . B3LYP/MWB52(Eu)/6-31 + G calculations [42] of the $[\text{Ln}(\text{H}_2\text{O})_6 \cdot \text{fumH}]^{2+}$ complex, with $\text{fumH}^- = \text{HOOCCH}=\text{CHCOO}^-$, show a drastic polarization of the $\text{C}=\text{C}$ double bond. The atomic charges (q) in $\text{HOOC}(\text{C}(2)\text{H}=\text{C}(1)\text{HCOO}^-$ are $q_{\text{C}(1)} = -0.091$ and $q_{\text{C}(2)} = -0.107$, whereas these charges in the complex $[\text{HOOC}(\text{C}(2)\text{H}=\text{C}(1)\text{HCOOLn}(\text{H}_2\text{O})_6)]^{2+}$ are $q_{\text{C}(1)} = -0.095$ and $q_{\text{C}(2)} = -0.066$, so that $\text{C}(2)$ becomes more electrophilic in the complex. The attack of water on the $\text{C}=\text{C}$ double bond also depends on the energy of the lowest unoccupied molecular orbital (LUMO), which decreases from $+0.0494 \text{ E}_h$ in fumH^- to -0.3147 E_h in $[\text{Ln}(\text{H}_2\text{O})_6 \cdot \text{fumH}]^{2+}$. The shapes of these orbitals are shown in Fig. 2, which clearly maintains its π -type symmetry in the complex. We also observe in Fig. 2 that the contribution from $\text{C}(2)$ to the LUMO appears larger in the complex than in the free species (fumH^-). This is indeed corroborated quantitatively by the sum of the square of the MO coefficients of $\text{C}(2)$ contributing to the LUMO, which increases from 0.0792 to 0.229 upon coordination. This suggests that the $[\text{Ln}(\text{H}_2\text{O})_6 \cdot \text{fumH}]^{2+}$ species is much more susceptible to the attack of water, particularly at $\text{C}(2)$.

The initial proposal of the decarboxylation/ CO_2 reductive coupling mechanism for the *in situ* formation of oxalate from pyridinedicarboxylic acids under hydrothermal conditions in the presence of $\text{Tb}(\text{NO}_3)_3 \cdot 5\text{H}_2\text{O}$ was based only on the observation that the reaction yield using pyridine-3,5-dicarboxylic acid increased from 42% to 82% in the presence of sodium oxalate [31]. However, the reaction yield was already quite high (70%) for pyridine-2,4-dicarboxylic acid [31] suggesting that the kinetics of oxalate formation is the relevant factor for the observed yields. In fact, addition of CO_2 sources such as Na_2CO_3 or NaHCO_3 did not affect the results [31] and this lack of activity was attributed to a speculatively low solubility of CO_2 produced from these sources under hydrothermal conditions [31]. We have also observed that dissolved CO_2 has no effect in the reaction yields of $[\text{Ln}_2(\text{fum})_2(\text{ox})(\text{H}_2\text{O})_4] \cdot 4\text{H}_2\text{O}$ ($\text{Ln} = \text{Eu}, \text{Gd}$, and Tb). A more detailed analysis of this *in situ* formation of oxalate was performed by a GC–MS study [32] of the solution resulting from the hydrothermal treatment of a mixture containing $\text{Sm}(\text{NO}_3)_3 \cdot 5\text{H}_2\text{O}$ (0.234 mmol), pyridine-3,5-dicarboxylic acid (0.239 mmol), 4,4'-bipyridine (0.230 mmol), and H_2O (5.0 mL, 0.278 mol) [32]. The analysis revealed molecular ion peaks at $m/e = 111, 156, 168$, and 194 , which were assigned to 3,5-dihydroxypyridine (pyridine-3,5-diol), 4,4'-bipyridine, protonated pyridine-3,5-dicarboxylic acid, respectively, and

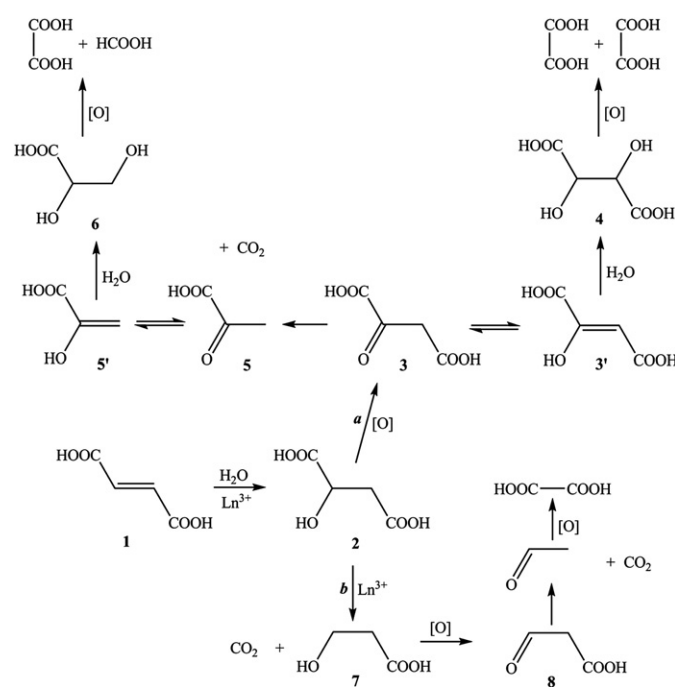
the largest m/e signal was not identified [32]. Notice that the signal $m/e = 111$ can also be assigned to pyridine-2,6-diol formed by hydroxylation of pyridine via Gillard mechanism [4,43]. Coordination of pyridine to the lanthanide ion could activate the α -position for hydroxylation and this step can be prior or simultaneous to the decarboxylation one. This study [32] is the only one to speculate about the electron source for the reductive coupling of CO_2 from decarboxylation, namely, a divalent lanthanide ion, Ln^{2+} . This is improbable because the standard potentials of $\text{Ln}^{3+} + e^- \rightarrow \text{Ln}^{2+}$ are -0.35 , -1.55 , -2.6 , -2.9 , -3.7 , and -3.9 V for $\text{Ln} = \text{Eu}$, Sm , Dy , Ho , Tb , and Gd , respectively [7], so they would hardly be produced in aqueous media even under hydrothermal conditions. Indeed, compounds with Eu^{2+} are available even in protic solvents, however, reducing agents containing Yb^{2+} and Sm^{2+} require specialized synthesis and conditions (mainly, inert solvent), whereas it was only recently that Nd^{2+} , Dy^{2+} , Tm^{2+} , and La^{2+} compounds and complexes have been prepared under very controlled and special conditions [44]. Given the special care and handling that divalent lanthanide requires it is unlikely that they would be present under hydrothermal conditions. In addition, divalent lanthanides present widely distinct reactivities [44] as can be inferred by the reduction potentials presented above. So, oxalate formation and thus reaction yields of lanthanide-oxalato solids should be strongly dependent upon the nature of the lanthanide. However, in all synthesis of lanthanide-oxalato based MOFs with *in situ* formation of oxalate under hydrothermal conditions the reaction yields have no correlation with redox potentials of Ln^{3+} to Ln^{2+} . For example, the $[\text{Ln}_2(\text{fum})_2(\text{ox})(\text{H}_2\text{O})_4] \cdot 4\text{H}_2\text{O}$ frameworks were prepared in 70%, 65%, and 60% yields for Eu , Gd , and Tb , respectively, whereas the following yields [32] were obtained for synthesizing the solids $[\text{Ln}_2(3,5\text{-pdc})_2(\text{C}_2\text{O}_4)(\text{H}_2\text{O})_4] \cdot 2\text{H}_2\text{O}$: $\text{Eu} = 35\%$, $\text{Sm} = 44\%$, $\text{Ho} = 39\%$, and $\text{Dy} = 41\%$. Likewise, $[\text{Ln}(3,5\text{-pdc})-(\text{C}_2\text{O}_4)_{0.5}(\text{H}_2\text{O})_2] \cdot \text{H}_2\text{O}$ MOFs were obtained with yields: Lu 25%, Gd 23%, Tm , Yb , Dy , Sm , Ho , Eu , La , Nd , and Er ca. 20% [33]. All these results show that the reaction yields are independent of the lanthanide ion. It is also unclear how the divalent lanthanide species involved in this mechanism derives from Ln^{3+} , that is, which is the electron source capable of reducing these ions under the experimental conditions. Considering that all *in situ* oxalate formation grouped under red4) employs nitric acid or sulfuric acid and *N*-heterocycle carboxylic acids, it is more likely that oxalate is formed from hydrolysis-oxidation of C–N and C=C bonds mediated by the Ln^{3+} ions [15–18,20,45].

Indeed, it is well established that nitric acid can oxidize alcohols, ketones, and olefins to (poly)carboxylic acids [46–49]. For instance, the oxidation of cyclohexanol by nitric acid gives adipic acid (hexanedioic acid) in high yields as well as glutaric (pentanedioic), succinic (butanedioic), and oxalic (ethanedioic) acids even in the absence of vanadium catalyst [46,48]. Analysis of the gases produced in these oxidation reactions indicated the presence of a large amount of N_2O as well as NO , NO_2 , CO_2 and N_2 [46]. We propose a mechanism (Scheme S1 in Supplementary Materials) for the *in situ* formation of oxalic acid from fumaric acid under hydrothermal conditions in the presence of Ln^{3+} and nitrate using only well-known reactions [46–50]. This mechanism involves nitro- and nitroso-intermediates obtained from common organic reactions (nitration and nitrosation) that occur in nitric acid aqueous medium [50] and is based on the oxidation mechanism of cyclohexanol by nitric acid [48]. These intermediates are suitable for acid catalyzed hydrolysis to form one or two mols of oxalic acid, depending upon the pathway. Nitrosonium ion is usually generated from nitrite in acid media as $\text{NO}_2^- + 2\text{H}^+ \rightarrow \text{NO}^+ + \text{H}_2\text{O}$ or from nitrogen tetroxide, initially formed by the first trace of oxidation, which ionizes in nitric acid to nitrosonium nitrate, namely, $\text{N}_2\text{O}_4 \rightarrow \text{NO}^+ + \text{NO}_3^-$ [47].

In order to ascertain if these nitro- or nitroso-intermediates are involved in the formation of oxalic acid we employed other oxidants instead of nitric acid. The same hydrothermal treatment was performed in 5 mL of aqueous solutions containing europium chloride (EuCl_3 , 0.4 mmol), fumaric acid ($\text{C}_4\text{H}_4\text{O}_4$, 1.5 mmol) and sodium hypochlorite (NaOCl , 1.2 mmol) or hydrogen peroxide (H_2O_2 , 1.2 mmol) or sodium

sulfate (Na_2SO_4 , 1.2 mmol) as oxidant (Figs. S10–S15). The first experiment afforded **1Eu** in 41% yield based on the europium, according to XPDR (Fig. 1b), whereas the latter experiment yielded a white precipitate that is clearly not **1Eu** (Fig. 1c). No solids were observed in the experiment with hydrogen peroxide. Also, all solids incorporated Eu^{3+} because they showed its typical luminescence under UV excitation. The conditions with sodium sulfate are being optimized to afford single crystals suitable for structural characterization. In addition, the aqueous solution of fumaric acid, europium chloride with the sodium hypochlorite oxidant was left at room temperature and atmospheric pressure for 3 days and led to the formation of Eu^{3+} containing solid (54% yield) that is not **1Eu** (Fig. 1d). Considering that oxalate was formed only under hydrothermal condition in the presence of strong oxidant (nitrate, NaOCl) we propose two multistep mechanisms (**a** and **b**), depicted in Scheme 1, which can account for the main observed results.

Mechanism **a** involves only hydration, oxidation and decarboxylation steps of activated substrates [50]. Whereas, mechanism **b** requires hydro(solvo)thermal conditions similar to those for oxidation of ethanol to oxalic acid [27]. Both mechanisms involve a hydration step of fumaric acid mediated by Ln^{3+} forming 2-hydroxysuccinic acid **2**. In mechanism **a**, oxidations of α,β -hydroxydiacids **2** and α,β -dihydroxydiacids **4** to the corresponding α,β -ketodiacids **4** (tartaric acid) and tetraacids (oxalic acid) are quite feasible due to the activation by the carboxylic groups [50]. Note that oxidative cleavage of C–C bond in 2,3-dihydroxysuccinic acid **4** (tartaric acid) is known for a long time and occurs even in mild conditions [51]. Also, the intermediate hydration step is quite viable due to the activation by the carboxylic groups in **3'** and by the presence of Ln^{3+} that might mediate this step. Alternatively, the β -ketodiacid (2-oxosuccinic acid **3**) can be promptly decarboxylated via a cyclic six-center mechanism [50] to give 2-oxopropanoic acid **5** that can be hydrated to form 2,3-dihydroxypropanoic acid **6** (glyceric acid), which also undergoes typical oxidation reactions to give oxalic acid and formic acid. Note that mechanism **a** produces 2 mol of oxalic acid for each mol of fumaric acid via intermediate **3** (entries 1–4 in Table 1) or 1 mol of oxalic acid when a decarboxylation step (intermediate **5**) is present (entries 5–



Scheme 1. Main reaction pathways proposed for oxalic acid formation from fumaric acid in low pH aqueous solution of trivalent lanthanide ions and oxidant (nitric acid or sodium hypochlorite) under hydrothermal conditions.

Table 1

Global reactions involved in mechanisms **a** and **b** with the corresponding standard reducing potential, E^0 in V, of the oxidation step.

Entry	Global reaction	E^0 (V)
Mechanism a via intermediate 3		
1	$C_4H_4O_4 + 4HNO_3 \rightarrow 2C_2H_2O_4 + 4HNO_2$	+0.93
2	$C_4H_4O_4 + 4HClO \rightarrow 2C_2H_2O_4 + 4HCl$	+1.48
3	$C_4H_4O_4 + 4H_2O_2 \rightarrow 2C_2H_2O_4 + 4H_2O$	+1.78
4	$C_4H_4O_4 + 4H_2SO_4 \rightarrow 2C_2H_2O_4 + 4H_2SO_3$	+0.17
Mechanism a via intermediate 5 (decarboxylation)		
5	$C_4H_4O_4 + 4HNO_3 \rightarrow C_2H_2O_4 + CH_2O_2 + CO_2 + 4HNO_2$	+0.93
6	$C_4H_4O_4 + 4HClO \rightarrow C_2H_2O_4 + CH_2O_2 + CO_2 + 4HCl$	+0.93
7	$C_4H_4O_4 + 4H_2O_2 \rightarrow C_2H_2O_4 + CH_2O_2 + CO_2 + 4H_2O$	+1.78
8	$C_4H_4O_4 + 4H_2SO_4 \rightarrow C_2H_2O_4 + CH_2O_2 + CO_2 + 4H_2SO_3$	+0.17
Mechanism b		
9	$C_4H_4O_4 + 5HNO_3 \rightarrow C_2H_2O_4 + 2CO_2 + 5HNO_2 + H_2O$	+0.93
10	$C_4H_4O_4 + 5HClO \rightarrow C_2H_2O_4 + 2CO_2 + 5HCl + H_2O$	+1.48
11	$C_4H_4O_4 + 5H_2O_2 \rightarrow C_2H_2O_4 + 2CO_2 + 6H_2O$	+1.78
12	$C_4H_4O_4 + 5H_2SO_4 \rightarrow C_2H_2O_4 + 2CO_2 + 5H_2SO_3 + H_2O$	+0.17

8 in Table 1). In mechanism **b**, the 2-hydroxysuccinic acid **2** intermediate undergoes α -hydroxydecarboxylation also mediated by Ln^{3+} to produce 3-hydroxypropionic acid **7**, which was invoked to explain the formation of 3-(pyridine-4-carboxylic acid)propionic acid [36]. We have assumed that Ln^{3+} mediates the decarboxylation of **2** since the release of CO_2 from α -hydroxyacids is not very common. The β -hydroxyacid intermediate (3-hydroxypropionic acid **7**) can be oxidized to β -ketoacid (3-oxopropanoic acid **8**) that undergoes decarboxylation via a cyclic six-center mechanism [50] to form acetaldehyde, which is further oxidized to oxalic acid. Alternatively, the β -hydroxyacid **7** can undergoes a decarboxylation, probably mediated by Ln^{3+} , to yield ethanol that is known to produce oxalic acid by oxidation under hydro(solvo)thermal conditions [27] as commented before (see entry ox5).

One relevant feature of the proposed mechanism **a** is that its pathways are based upon known reactions and feasible intermediates that require no special conditions besides aqueous acidic media and Ln^{3+} mediation. Mechanism **b** is also based on known reactions, despite the mechanism for the oxidation of ethanol under hydro(solvo)thermal conditions [27] is still undetermined. From these mechanisms, at least two driving forces for producing lanthanide-oxalato MOFs can be identified. Firstly, the low solubility of the frameworks in aqueous media introduces a dual role for the trivalent lanthanide ions in addition to the mediation of hydration/decarboxylation processes. Secondly, the oxidation reactions with nitric acid, sodium hypochlorite and hydrogen peroxide are highly exergonic processes as seen in Table 1, which can compensate for the energy demand of oxalic acid formation. The low redox potential (see Table 1) associated with sodium sulfate can explain the failure of this oxidant in producing oxalic acid. In addition, the coordinating ability of sulfate ions could account for the production of insoluble frameworks with lanthanide ions. We dissolved the solid obtained with sodium sulfate in HCl(aq). Preliminary NMR analysis has not shown oxalic acid in the resulting solution. Regarding the result with hydrogen peroxide, we propose that it is a too strong oxidant (see entries 3, 7, and 11 in Table 1) that is unable to selectively produce oxalic acid under hydrothermal conditions. In fact, ^{13}C NMR analysis of the resulting solution shows several signals of carbonyl compounds that are not oxalic acid (Figs. S13–S14). This shows the importance of choosing an appropriate oxidant that can selectively yield oxalate at a rate compatible to the formation of the framework.

Two important corroborating evidences for the mechanisms were the detection of nitrite and formic acid in the solutions resulting from the hydrothermal treatments. Nitrite was detected [52] only in the solutions where the starting materials contained nitrate, Ln^{3+} and fumaric

acid. NMR analyses (Figs. S10 and S14) performed before and after addition of formic acid provided evidence for its presence in the solutions where **1Eu** was formed (nitrate or sodium hypochlorite).

Another interesting feature of the mechanisms in Scheme 1 is that they can be generalized to explain the formation of oxalic acid from *N*-heterocyclic compounds by decarboxylation, hydroxylation, tautomerization, hydration and oxidative cleavage reactions promoted by Ln^{3+} ions and by oxidants such as nitric acid or perchloric acid ($HClO_4 + 2H^+ + 2e^- \rightarrow HClO_3 + H_2O$, $E^0 = +1.19$ V). These reactions could explain most of lanthanide-oxalato coordination polymers and MOFs formed under hydrothermal conditions grouped in entry red4 listed above. However, a detailed mechanism for these *in situ* oxalate formations is beyond the scope of the present contribution.

Finally, NMR analyses (Figs. S6, S8 and S9) of the resulting aqueous solution from hydrothermal treatment revealed the presence of propan-2-ol (isopropanol) [53] when nitrate was used as an oxidant. NMR analyses of the initial solution showed no traces of propan-2-ol, so it can be assumed that it was formed during the hydrothermal treatment. At this moment we are unable to account for the formation of propan-2-ol.

From Fig. 1 it can be observed that the crystallinity of the solids is quite dependent upon reaction conditions, such as, strength of the oxidant and temperature. Thus, it might also be inferred that the morphology of the solids could be controlled by the reaction conditions so a new approach for obtaining MOFs and coordination polymers with desired nano- and microparticle structures [54] could be developed.

In conclusion, we obtained $[Ln_2(fum)_2(ox)(H_2O)_4] \cdot 4H_2O$ ($Ln = Eu$ **1Eu**, Gd **1Gd**, and Tb **1Tb**) single crystals in good yields from hydrothermal treatment of aqueous acid solutions of trivalent lanthanide ions, fumaric acid and nitrate as oxidant, with oxalate being formed *in situ*. These compounds were additionally characterized by infrared spectroscopy (Fig. S16), thermogravimetric analyses (Fig. S17), excitation and emission spectra (Figs. S18 and S19 for **1Tb**), and phosphorescence spectrum of **1Gd** (Fig. S20). The excitation and emission spectra for **1Eu** and **1Tb** were already fully discussed [5] and they agree quite well with the available spectra for the “as-prepared” or “re-hydrated” samples [5]. Employing sodium hypochlorite instead of nitrate also afforded **1Eu** in powdered form under hydrothermal conditions. Other oxidants are not strong enough (e.g., sodium sulfate) to form oxalic acid or too strong (e.g., hydrogen peroxide) to selectively yield oxalic acid. Also, hydrothermal conditions are needed to form oxalate *in situ*. The mechanisms proposed are based upon well-known reactions (hydration, oxidation, and decarboxylation) in acid aqueous solutions. These mechanisms explain the role of the trivalent lanthanide ions as well as the fact that an oxidant such as nitrate or hypochlorite is indispensable for obtaining the MOF. It might also be generalized to give an alternative explanation for the *in situ* formation of oxalic acid during hydrothermal treatment of acid aqueous solution containing trivalent lanthanides, *N*-heterocyclic compounds and some oxidant such as nitrate or perchlorate.

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Appendix A. Supplementary material

Crystal data for **1Gd**: CCDC 859852 and the supplementary crystallographic data. 1H and ^{13}C NMR spectra used qualitative and

quantitative analysis and Scheme S1 for the reaction pathways involving nitroso- and nitro-intermediates. Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.inoche.2012.04.036>.

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